ECLAIR: Anytime-Valid Screening of LLM-Generated Constraint Models, with ε-Certification of Optimum-Value Agreement

Quang-Vinh Dang[1*]

[1*]British University Vietnam, Hung Yen, Vietnam.

Corresponding author(s). E-mail(s): vinh.dq4@buv.edu.vn;

Abstract

Large language models translate natural-language problem descriptions into formal constraint models, but they lack the mathematical guarantees required for formal correctness, and no ground truth exists against which a generated model can be checked — the oracle problem. Existing verification approaches are detectors: they report violations and empirical detection rates, but give no statistical error control, no principled stopping rule, and no way to choose the next test under a compute budget. We introduce ECLAIR, a screen-then-certify layer with two guarantees pointing in opposite directions. For *screening*, each candidate carries an e-process betting against its faithfulness, fed by three tiers of imperfect probes (exhaustively enumerated micro-instance oracles, provably valid metamorphic relations, cross-model majority checks); Ville's inequality bounds by $\boldsymbol{\alpha}$ the probability of ever falsely rejecting a faithful candidate, under optional stopping and adaptive Kelly-style budget routing, and e-BH controls the false discovery rate of the rejected set. For *acceptance*, a second, opposite-direction e-process on fresh oracle instances $\boldsymbol{\varepsilon}$-certifies the surviving candidate: the probability of ever certifying a model whose optimum-value disagreement rate with the specification exceeds $\boldsymbol{\varepsilon}$, on a declared exhaustively enumerable micro-instance distribution, is at most $\boldsymbol{\alpha}$ — and that scope is the certificate's limit, covering neither feasible-set equivalence nor behaviour at deployment scale. On a mutation testbed of five problem families with real CP-SAT probes, no faithful candidate was ever falsely rejected, **89–96%** of injected faults were detected, and an independent audit of every issued certificate — **400** fresh instances per certified model — places all **289** error intervals entirely below $\boldsymbol{\varepsilon}$, with none above and none inconclusive.

Keywords: constraint modelling, large language models, e-processes, anytime-valid inference, solver-in-the-loop verification, false discovery rate

1 Introduction

Constraint programming's long-standing adoption bottleneck is the expertise barrier: translating a problem description held in natural language into a formal model. Large language models (LLMs) now cross that barrier routinely — generating CPMpy, MiniZinc, or MILP models from plain-English specifications [1–3] — but they do so without any intrinsic guarantee of correctness. A generated model that compiles and solves is not necessarily the model that was asked for: a misread coefficient, a dropped constraint family, or a flipped inequality produces a plausible artifact that returns confidently wrong optima. Because the specification exists only in natural language, there is no oracle to compare against; this is the central verification difficulty the field currently faces.

The question a practitioner asks of any verification layer is: *what guarantee do I get on the model I finally accept?* Existing answers are heuristic. Detection frameworks based on metamorphic relations or mutation-style testing report empirical detection rates, but offer no control over the probability of certifying an unfaithful model, no principled rule for when enough testing has been done, and no guidance on which of many possible tests — differing wildly in cost and power — to run next under a finite solver budget. Fixed-sample statistical tools do not repair this: a generate–test–repair loop inherently peeks at accumulating evidence and adapts, which invalidates fixed-sample p-values and split-conformal calibration alike.

A concrete miniature fixes ideas. A dispatcher asks for "a knapsack model: pick items within the truck's 120 kg limit, maximizing value; items 3 and 7 cannot travel together." A language model returns clean CPMpy code — but read `120` as the *count* bound of a different template, wired the weight vector where values belong, or emitted `sum(w*x) >= cap` after seeing a covering example in its context. Each of these artifacts compiles, solves in milliseconds, and returns a confident number. Asking a second model and comparing helps only if their errors differ; testing on a hand-checkable toy instance helps only if the error manifests there; and the practitioner who keeps testing "until it looks fine" has silently destroyed the error control of whatever test they were using. What is missing is an accounting system for the accumulated evidence — one that survives the practitioner's own adaptivity.

This paper introduces **ECLAIR** (E-process Certification of LLM-generated constraint models with Adaptive Instance Routing), a certification layer built on the game-theoretic statistics of e-processes [4–6]. The design follows three observations.

First, *optional stopping is the workflow.* E-processes are designed for continuous monitoring: rejecting whenever the wealth process crosses $1/\alpha$ controls the type-I error at α uniformly over all stopping times (Ville's inequality [7]), so the loop may stop the moment evidence suffices — a direct compute saving — and may resume testing at any time without penalty.

Second, *probes are heterogeneous, imperfect, and dependent.* A brute-forced micro-instance is a near-perfect oracle but expensive; a metamorphic relation is provably false-alarm-free but weak; a cross-model agreement check is cheap, noisy, and entangled with the candidate pool. E-values multiply across adaptively chosen, dependent probes as long as each factor is a valid e-variable conditionally on the past, and validity under the composite null "this candidate is faithful" needs only an upper *bound* on each probe family's null alarm rate rather than its exact null distribution — provided that bound holds for every conditional alarm probability the deployed system encounters. For our exact micro-instance oracle and proved metamorphic relations that requirement is discharged by proof; for the pool-entangled cross-model tier it is a deployment assumption we state, stress-test, and do not claim to have established (Section 5).

Third, *the budget question is a portfolio question*. Choosing which (candidate, probe) pair to run next to maximize expected log-evidence growth per probe-second is Kelly-*style* evidence acquisition in the spirit of the GROW criterion [6, 8] — a plug-in heuristic, not a policy we prove growth-optimal for the deployed environment (Section 5); because the routing decision is measurable with respect to the past, it affects power only, never validity.

Contributions.

(i) A formal screen-then-certify protocol for LLM-generated constraint models: per-candidate falsification e-processes over a three-tier probe pool, with hard certificates (probes whose false-alarm probability is exactly zero by proof) embedded as infinite e-values (Section 4). (ii) Guarantees in both directions, kept carefully apart: anytime-valid false-rejection control for the screening side under arbitrary predictable routing (Theorem 2, with conservative calibration preserving the supermartingale property, Lemma 1, unconditionally for the proved tiers and conditionally on a stated deployment assumption for the pool-entangled one); false discovery rate control of the rejected set via e-BH [9] (Proposition 4); a budget bound on the expected probe time to reject an unfaithful candidate (Proposition 5); and — the guarantee on the model one finally accepts — an ε-certification theorem (Theorem 3): a second, opposite-direction e-process on fresh oracle instances bounds by α the probability of ever certifying a candidate whose disagreement rate with the specification is at least ε, robustly to data-dependent selection of the certified candidate. (iii) A verified probe library for five problem families — knapsack with conflicts, set cover, graph colouring, generalized assignment, and on-time job selection — including per-family metamorphic relations proved valid at specification level (Proposition 7). (iv) An open implementation on CPMpy [10] and OR-Tools CP-SAT [11] and a five-part empirical study (Section 7): a mutation testbed in which validity holds without an observed exception at every nominal level while 89–96% of injected faults are caught; ablations quantifying the contribution of each probe tier, including an empirical demonstration that pool-entangled probes consume the conservative validity margin; a fair-design baseline study on identical pools in which, with probes and e-factors held literally identical, sequential accounting strictly contains the fixed-sample rejection set while consuming one third of the probes, and the practitioner's no-rejection alternatives (single-shot, self-consistency voting) hand over an unfaithful model 19–54% of the time with no level to quote; an end-to-end smoke test with two LLM families on real generated models; and an ambiguity study whose honest negative result — current models normalize boundary-language ambiguity to the conventional reading — sharpens where the silent-error regime actually lives.

2 Related work

LLMs for constraint modelling.

In-context learning pipelines translate natural-language descriptions into CPMpy models with retrieval and self-refinement [1]; benchmark suites evaluate such systems on CSPLib-derived corpora [12]. Agentic systems iterate modelling with solver feedback: CP-Agent [3] maintains a persistent Python session around CPMpy, and MCP-Solver [13] exposes constraint solvers to LLMs through a model context protocol. ConstraintLLM [2] targets industrial-level CP with a neuro-symbolic architecture. All these systems *generate*; ECLAIR is complementary — it decides, with a guarantee, whether to *trust* what was generated, and can wrap any of them.

Testing lineage.

Two software-testing traditions supply ECLAIR's probe vocabulary. Metamorphic testing [14] addresses the oracle problem by checking necessary relations between outputs on related inputs — exactly our tier B, with the strengthening that relations proved against the *specification* of an optimization problem have null alarm probability zero, turning a testing heuristic into a hard certificate inside a statistical process. Mutation testing [15, 16] contributes the evaluation methodology: systematically injected faults with screened non-equivalence are the standard way to measure a test suite's power when real faults are scarce, and our functional-corruption testbed (Section 7) is its adaptation to model faithfulness — with the twist that here the "test suite" carries a statistical guarantee, so the mutation score is a *power* estimate under a controlled *level*, not a standalone adequacy number.

Verifying generated models.

Two recent lines attack verification directly. Falsification-based verification [17] derives sound test batteries from LP duality, monotone comparative statics, and Le Chatelier-type principles, in which a violated relation certifies unfaithfulness with zero false-positive rate by construction; agent-based validation [18] generates a problem-level testing API, composes test cases, and measures adequacy by optimization-oriented mutation coverage. Both are detectors: they output violations and empirical detection rates. ECLAIR subsumes their binary verdicts as the special case of an infinite e-value and adds the statistical layer they lack — an anytime-valid level for the falsification side, an explicit acceptance guarantee for the certification side (Theorem 3), a stopping rule, and budget-aware test selection.

Anytime-valid inference.

E-values and e-processes provide measures of evidence valid at arbitrary stopping times [4, 5, 19]; safe tests and the GROW criterion connect optimal evidence growth to log-optimal (Kelly) betting [6, 8]; betting-based confidence sequences supply the adaptive wagering machinery [20]; and the e-BH procedure controls FDR under arbitrary dependence [9]. ECLAIR ports this toolkit into constraint-model verification, where, to our knowledge, sequential anytime-valid certification has not been attempted.

Why not fixed-sample or conformal tools.

It is worth being precise about why the natural statistical alternatives do not fit this workflow, since the certification problem looks superficially like ordinary hypothesis testing or distribution-free calibration. A fixed-sample test requires the number and identity of probes to be fixed before any outcome is seen; a generate–test–repair loop violates this at every step — it looks at the running evidence to decide whether to continue testing, which probe to run next, whether to repair, and when to stop, and each of those choices invalidates a fixed-sample p-value in the familiar optional-stopping way. Split-conformal calibration instead requires exchangeability between the calibration data and the test stream; here the "test stream" is adaptively generated (repaired candidates are chosen *because* of earlier outcomes; probe instances are routed by an adaptive policy), so exchangeability fails by construction. E-processes are designed for exactly this regime: validity is preserved under any data-dependent stopping, continuation, and selection rule, because the guarantee is a supermartingale property rather than a sampling-distribution property [4]. The practical consequence is the one that matters operationally: the certifier may spend its budget *where the evidence points*, stop the moment a decision is supported, and still quote the same α.

3 The certification problem

A specification s is a natural-language description of a parameterized combinatorial optimization problem, together with an instance distribution $\mathcal{D}$ over parameter values. A *candidate* m is an executable function mapping an instance $I \sim \mathcal{D}$ to a formal model $m(I)$ whose exact optimum we write $v_m(I) \in \mathbb{Z} \cup \{\text{INFEAS}\}$. The (unknown) faithful value $v^*(I)$ is the optimum of the problem s actually describes. Candidate m is *faithful* if $v_m(I) = v^*(I)$ for $\mathcal{D}$-almost-every I.

Given candidates $m_1, \ldots, m_k$ produced by an LLM ensemble, a compute budget B (probe-seconds), and a risk level α, the certifier must decide which candidates to reject, which (if any) to certify, or whether to abstain — without access to v^*. The null hypothesis attached to each candidate is

$$H_0(m): \quad m \text{ is faithful.}$$

$H_0(m)$ is *composite*: faithfulness does not determine the law of any probe outcome, which also depends on the instance generator, on checker noise, and (for cross-model probes) on the composition of the surviving pool. All guarantees below therefore hold uniformly over the null class, which we make precise through conditional bounds on probe alarm rates.

4 The ECLAIR framework

Figure 1 shows the architecture. An LLM ensemble proposes candidates; a cheap intake gate discards code that cannot build and solve a toy instance; every surviving candidate carries its own e-process; a router spends the solver budget one probe at a time; and three exits are possible — rejection (with a counterexample that can seed

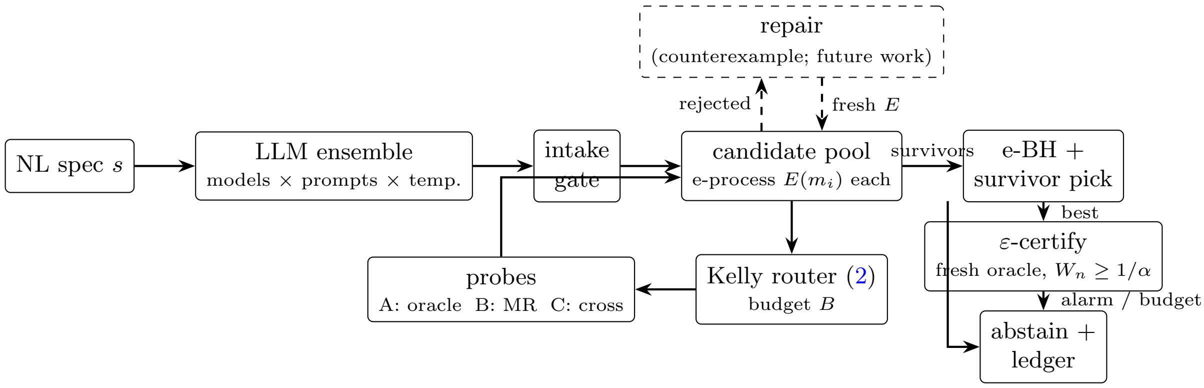

Fig. 1 ECLAIR's architecture. Every candidate carries its own falsification wealth process; the router spends the shared screening budget one probe at a time; terminal e-values enter e-BH; the single best survivor then faces the separate ε-certification e-process on fresh oracle instances — the only path to an accepted model — and any alarm or budget exhaustion there routes to abstention with the evidence ledger. The repair loop (dashed) is a natural extension this paper deliberately does not claim (Section 4.4)

a repaired candidate), ε-certification of the single best survivor by a separate fresh-oracle e-process, or abstention with the full evidence ledger. The subsections follow the data flow.

4.1 Probe tiers

A *probe* is a randomized procedure that consumes probe time and returns a binary alarm $X \in \{0, 1\}$ against a designated candidate. ECLAIR uses three tiers.

Tier A: micro-instance oracles.

Draw an instance small enough for exhaustive enumeration of the specification's decision space using an executable feasibility checker, and alarm iff the candidate's exact optimum differs from the enumerated optimum. With an exact checker the null alarm probability is zero; in general the checker itself is imperfect (in a fully LLM-generated pipeline it is LLM-compiled and validated by cross-agreement), so tier A is treated as a calibrated noisy probe.

Tier B: metamorphic relations (hard certificates).

A metamorphic relation (MR) transforms an instance I into I' such that the specification provably implies a relation between $v^*(I)$ and $v^*(I')$ — e.g. enlarging a knapsack capacity cannot decrease the optimum, deleting a graph edge cannot increase the chromatic number, consistently permuting item data leaves the optimum unchanged. The probe solves the *candidate* on both sides exactly and alarms iff the relation is violated. For a faithful candidate the alarm probability is exactly zero: both solves return true optima of instances of the true specification, and the relation holds for the specification by proof. A tier-B alarm is therefore a *hard certificate* of unfaithfulness.

Tier C: cross-model majority checks.

Solve the designated candidate and up to three surviving partners on a fresh mid-size instance (beyond enumeration reach — this tier's purpose); alarm iff at least two partners agree on a value and the candidate deviates from that majority. Tier C is cheap and noisy, and its null alarm rate depends on the pool: a faithful candidate alarms when a wrong majority forms. This entanglement is handled by calibration (Section 4.2) and is quantified empirically in Section 7.7.

4.2 Betting and conservative calibration

Let $\mathcal{F}_{t-1}$ denote everything observed before the t-th probe. For a tier-B probe the e-factor is $e_t = 1$ on pass and $e_t = \infty$ on alarm. For tiers A and C, ECLAIR bets a Bernoulli likelihood ratio

$$e_t \;=\; \left(\tfrac{p_1}{\bar{p}_0}\right)^{X_t} \left(\tfrac{1-p_1}{1-\bar{p}_0}\right)^{1-X_t}, \tag{1}$$

where $\bar{p}_0$ is a *conservative* (upper confidence) bound on the tier's null alarm rate and $p_1 \in (\bar{p}_0, 1)$ a working detection rate; both come from a held-out calibration corpus of known-faithful models and screened mutants, with $\bar{p}_0$ the Clopper–Pearson upper bound [21] at confidence 97.5%. Only $\bar{p}_0$ matters for validity; p_1 tunes power.

Lemma 1 (Conservative bets are safe) *Let $X \in \{0,1\}$ satisfy $\Pr(X = 1 \mid \mathcal{F}_{t-1}) \le \bar{p}_0$ almost surely under H_0, and let e be given by (1) with $0 < \bar{p}_0 < p_1 < 1$. Then $\mathbb{E}[\, e \mid \mathcal{F}_{t-1}\,] \le 1$ under H_0.*

Proof With $q = \Pr(X{=}1 \mid \mathcal{F}_{t-1})$, $\mathbb{E}[e \mid \mathcal{F}_{t-1}] = q\,\frac{p_1}{\bar{p}_0} + (1-q)\,\frac{1-p_1}{1-\bar{p}_0}$, which is affine and increasing in q because $p_1/\bar{p}_0 > 1 > (1-p_1)/(1-\bar{p}_0)$. At $q = \bar{p}_0$ it equals $p_1 + (1-p_1) = 1$; hence for $q \le \bar{p}_0$ it is at most 1. □

The candidate's wealth process is the running product $E_t(m) = \prod_{s\le t,\, s\to m} e_s$ over the probes routed to m, with $E_0(m) = 1$.

The betting arithmetic, numerically.

With the calibrated values of Section 7 (Table 3), the multipliers of (1) are concrete: a tier-A alarm multiplies wealth by $p_1/\bar{p}_0 = 0.473/0.0153 \approx 31$, a pass by $(1{-}p_1)/(1{-}\bar{p}_0) \approx 0.53$; a tier-C alarm multiplies by $\approx$3.0, a pass by $\approx$0.61; tier B multiplies by 1 on pass and rejects outright on alarm. Against the rejection threshold $1/\alpha = 20$ this means: a single tier-A alarm brings a fresh candidate to within one small step of rejection ($\log 31 > \log 20$ already, from wealth 1); two early tier-A passes push a candidate's wealth down to $\approx$0.28, after which the Kelly numerator of (2) for noisy tiers turns negative and the router parks the candidate on zero-bleed tier-B probes — exactly the flattening visible in Figure 2. The asymmetry (alarm evidence $\sim$31×, pass evidence $\sim$0.53×) is not a tuning choice but the likelihood-ratio geometry of a rare-under-null, common-under-alternative event.

4.3 Budget-aware routing

At each step ECLAIR selects

$$(m,p)^\star \;=\; \arg\max_{(m,p)} \frac{\mathbb{E}\big[\log e \;\big|\; q_m\big]}{c(p)}, \tag{2}$$

the candidate–probe pair maximizing expected log-wealth growth per probe-second — Kelly-*style* evidence acquisition in the spirit of the GROW criterion [6, 8], with the caveat of Section 5 that the plug-in posterior, working rates and mean costs make this a heuristic rather than a provably growth-optimal rule. Here q_m is a running (Bayes) posterior that m is unfaithful, maintained with the same calibrated rates and used *only* for routing; $c(p)$ is the tier's calibrated mean cost. For tiers A/C the numerator is $q_m G_1 + (1-q_m)G_0$ with G_1, G_0 the expected log-factors of (1) under alarm rates $p_1, \bar{p}_0$; for tier B it is $q_m\,\delta_B\big(\log(1/\alpha) - \log E_t(m)\big)$, the calibrated hard-certificate rate times the remaining log-gap to the rejection threshold. Because (2) is $\mathcal{F}_{t-1}$-measurable, routing affects power only — validity is untouched (Theorem 2).

4.4 Decisions: falsify, then certify, or abstain

The screening phase *falsifies*: candidate m is rejected the first time $E_t(m) \geq 1/\alpha$ (or on a tier-B alarm). When the screening budget is exhausted, the pool's terminal e-values enter the e-BH procedure [9] at level α, and the *survivors* are the candidates rejected by neither test. We are deliberate about what survival means: a survivor is *not falsified at evidence level* α — surviving is not a certificate, because the screening null is faithfulness and failing to reject a null is never evidence for it (Section 5).

The certificate comes from a second, opposite-direction e-process. The best survivor (smallest e-value) enters a *certification phase*: fresh tier-A oracle probes on i.i.d. micro instances, each pass multiplying a certification wealth by $1/(1-\varepsilon)$, each alarm hard-falsifying the candidate (the checker is exact). Crossing $1/\alpha$ declares the candidate *ε-certified*: its disagreement rate with the specification on the micro distribution is below ε, at anytime level α, robustly to how the candidate was selected (Theorem 3). Exactly one candidate per run is ever attempted — attempting a second at the same threshold would spend the level (Remark 1) — so if the attempted survivor falls, the run abstains from certification. If no candidate survives screening — or the attempt does not certify — ECLAIR *abstains* and returns the evidence ledger: a structured explanation of which probes killed which reading of the specification, which doubles as a signal for conversational re-elicitation. Repair (feeding a violating probe back to the LLM as a counterexample) is a natural extension of this loop that we deliberately do not claim: FDR control over adaptively grown pools needs online multiple-testing machinery beyond Proposition 4. Algorithm 1 summarizes.

4.5 A worked run

Figure 2 traces one screened pool — three faithful candidates and three screened mutants of the set-cover family — under Kelly routing at $\alpha = 0.05$ with a 286 ms screening budget of measured probe time. Every mechanism of the framework is visible

Algorithm 1 ECLAIR

Require: spec s, candidates $m_1{:}k$, budget B, level α, calibrated $(\bar{p}_0, p_1, \delta_B, c)$ per tier
$E(m_i) \leftarrow 1$, $q_{m_i} \leftarrow q_{\text{prior}}$ for all i
while $B > 0$ **and** some candidate alive **do**
  $(m, p) \leftarrow$ maximizer of (2) over alive candidates and tiers
  run probe p on m; observe alarm X, measured cost c; $B \leftarrow B - c$
  if p is tier B **and** $X = 1$ **then** reject m ▷ hard certificate
  else
    $E(m) \leftarrow E(m) \cdot e(X)$ by (1); update q_m by Bayes
    if $E(m) \geq 1/\alpha$ **then** reject m
    end if
  end if
end while
survivors $\leftarrow$ candidates rejected by neither Ville threshold nor e-BH at level α
$m^\star \leftarrow \arg\min_{\text{survivors}} E(m)$ ▷ certification: single attempt
while attempt budget remains **do**
  probe $m^\star$ on a fresh micro instance (tier A); $W \leftarrow W \cdot \frac{\mathbf{1}\{\text{pass}\}}{1-\varepsilon}$
  if $W \geq 1/\alpha$: **return** $m^\star$ as ε-certified + ledger
  if alarm: hard-falsify $m^\star$; **break**
end while
return abstain with the evidence ledger

in the paths. Two mutants are driven across the rejection threshold within 60 ms: the router's prior makes every candidate equally suspect at the start, and a single tier-A alarm already suffices from initial wealth ($\log 31 = 3.43 > \log 20 = 3.00$). The third mutant survives its first probes by luck, its posterior q_m drops, and the router deprioritizes it — until the pool thins, its relative score rises again, and a single tier-B hard certificate (the near-vertical jump at $\sim$290 ms) removes it. The faithful candidates' paths tell the validity story: early tier-A/C passes drive their wealth *down* (a pass multiplies by $(1 - p_1)/(1 - \bar{p}_0) < 1$), after which the router — seeing low q_m — feeds them almost exclusively zero-bleed tier-B probes, so their wealth flattens far below the threshold and the remaining budget is spent where Lemma 1 makes passes free. The surviving pick is the faithful candidate with the smallest terminal wealth — and in the full protocol it would now enter the certification phase of Section 4.4; its ledger records every probe shown in the figure. Table 1 lists the first fourteen ledger entries verbatim: the entire decision logic of the framework — likelihood ratios, threshold crossings, router hand-offs to the zero-bleed tier — is legible in fourteen rows of arithmetic.

5 Guarantees

Before stating guarantees we separate, deliberately, what is proved from what is assumed — and, more importantly, *which direction each guarantee points.* The falsification side (Theorem 2) controls the probability of ever rejecting a *faithful* candidate; it says nothing about survivors, and surviving is never evidence of faithfulness. The

Table 1 The first fourteen ledger entries of the run in Figure 2 (m_1–m_3 faithful, m_4–m_6 mutants; $\alpha = 0.05$, rejection at $\log E \geq \log 20 \approx 3.00$). Steps 4–5: one tier-A alarm each rejects m_4, m_5 outright ($\log 31.0 = 3.43$). The lucky mutant m_6 passes its first probes and is parked, with the faithful, on zero-bleed tier-B probes (steps 11–14) — it is rejected much later by a hard certificate (Fig. 2).

Step	Candidate	Tier	e-factor	$\log E$ after
1–3	m_1, m_2, m_3	A (pass)	0.535	-0.626
4	m_4	A (**alarm**)	31.0	3.435 $\rightarrow$ **reject**
5	m_5	A (**alarm**)	31.0	3.435 $\rightarrow$ **reject**
6	m_6	A (pass)	0.535	-0.626
7–10	m_1, m_2, m_3, m_6	A (pass)	0.535	-1.252
11–14	m_1, m_2, m_3, m_6	B (pass)	1.000	-1.252

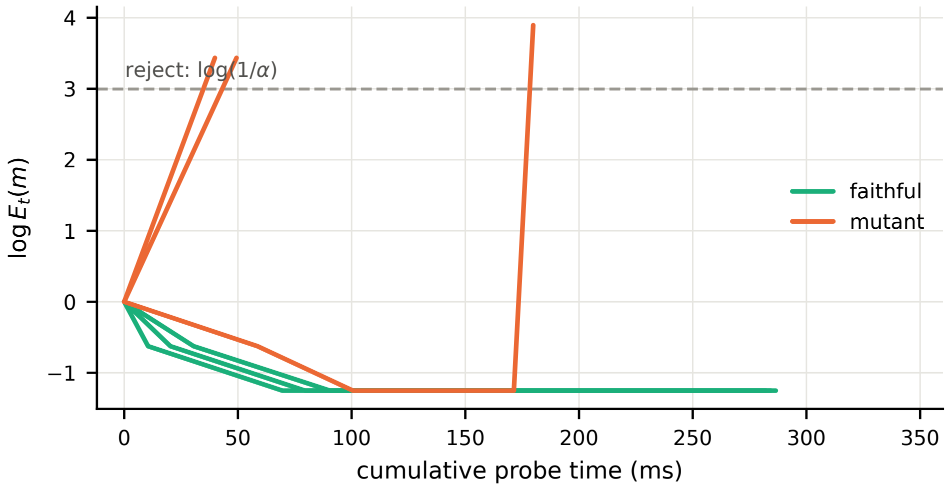

Fig. 2 Wealth trajectories $\log E_t(m)$ of one screened pool (three faithful set-cover candidates, three screened mutants; Kelly routing, $\alpha = 0.05$). Mutant paths cross the rejection threshold (dashed); the late near-vertical jump is a tier-B hard certificate. Faithful paths drift down on early noisy passes, then flatten as the router shifts them to zero-bleed tier-B probes

acceptance side (Theorem 3) is a separate, opposite-direction test that earns the word "certified."

Assumption 1 (Null alarm rates) Under $H_0(m)$: (a) tier-B probes alarm with probability zero; (b) tier-A probes alarm with probability zero conditionally on any history; (c) each tier-C probe routed to m alarms with conditional probability at most its calibrated bound $\bar{p}_0^C$, given $\mathcal{F}_{t-1}$.

Parts (a) and (b) are *proved*, not calibrated, in our instantiation: (a) is Proposition 7, and (b) holds by construction whenever the tier-A checker is exact — both solves are exact, so a faithful candidate's optimum equals the enumerated optimum on every instance, whatever the history. (Our experiments use hand-written independent checkers, so (b) is a theorem there; in a pipeline whose checkers are themselves LLM-generated, tier A degrades to a calibrated tier and moves into part (c)'s regime.) Part (c) is a genuine *deployment assumption*, and we are explicit about what our calibration does and does not deliver for it. The cross-model alarm event depends on the surviving pool, so (c) is a statement about *every* conditional alarm probability encountered at deployment time — across candidates, pool compositions, routing histories, and problem families. A Clopper–Pearson interval certifies coverage for a *single* binomial parameter under the calibration sampling design; it does not, and cannot, establish that uniform conditional bound. We therefore make no attempt to convert the confidence level into an unconditional guarantee: all results below are stated *conditionally on Assumption 1*, and the calibration corpus, the cross-family transfer tests (Section 7.9), and the tier ablation (Section 7.7) are *empirical stress tests* of part (c), not proofs of it. A calibration theorem giving uniform conditional coverage — for instance via stratified worst-case calibration over pool-composition buckets, or online recalibration with an explicit risk budget — would upgrade the conditional guarantee to an unconditional one; we regard this as the most important open problem the framework leaves (Section 10). Configurations that use only tiers A and B are exempt: for them Assumption 1 is a theorem given an exact checker, and the guarantees below are unconditional.

Theorem 2 (Anytime-valid falsification) Conditionally on Assumption 1*, for any predictable routing rule and any stopping rule,*

$$\Pr_{H_0(m)}\big(\exists\, t:\ E_t(m) \geq 1/\alpha\big) \ \leq\ \alpha.$$

The conditioning is not decorative: with tiers A and B only it is discharged by proof (exact checker, proved relations) and the bound is unconditional; with tier C active it rests on the deployment assumption above, which our experiments stress-test but do not establish. Candidates that are never rejected are not falsified at evidence level α*; this theorem attaches no positive certificate to them.*

Proof Each factor contributed to $E(m)$ is nonnegative and, by Lemma 1 (tiers A/C) or by the zero-alarm property (tier B, factor $\equiv 1$ a.s. under H_0), satisfies $\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] \leq 1$; predictability of the routing rule makes "which probe, on which candidate" a $\mathcal{F}_{t-1}$-measurable choice, so the tower property gives $\mathbb{E}[E_t(m)] \leq 1$ and $E_t(m)$ is a nonnegative supermartingale. Ville's inequality [7] concludes. □

Theorem 3 (ε-certification of the accepted model) *Fix $\varepsilon \in (0,1)$ and define the operational unfaithfulness null for a candidate m,*

$$H_0'(m): \quad \mathrm{err}(m) := \Pr_{I\sim\mathcal{D}_\mu}\big(v_m(I) \neq v^*(I)\big) \ \geq\ \varepsilon,$$

where $\mathcal{D}_\mu$ is the micro-instance distribution and the tier-A checker is exact. Let a candidate — selected by any *rule measurable with respect to the screening phase — be probed on fresh i.i.d. instances $I_1, I_2, \ldots$ with alarms $X_j = \mathbf{1}\{v_m(I_j) \neq v^*(I_j)\}$, and let $W_n = \prod_{j \le n} \frac{\mathbf{1}\{X_j=0\}}{1-\varepsilon}$. Declare m ε-*certified *when $W_n \ge 1/\alpha$, i.e. after $n \ge \lceil \log(1/\alpha)/\log \frac{1}{1-\varepsilon} \rceil$ consecutive passes, where* exactly one *candidate per run — the best survivor — is ever attempted. Then*

$$\Pr\big(\textit{the attempted } m \textit{ has } \mathrm{err}(m) \ge \varepsilon \textit{ and is } \varepsilon\textit{-certified}\big) \;\le\; \alpha.$$

Proof Under $H_0'(m)$, $\Pr(X_j = 1 \mid \text{past}) \ge \varepsilon$ because the I_j are fresh draws from $\mathcal{D}_\mu$, independent of the screening data and of earlier certification probes; hence $\mathbb{E}[\mathbf{1}\{X_j = 0\}/(1-\varepsilon) \mid \text{past}] = (1 - \Pr(X_j = 1 \mid \text{past}))/(1-\varepsilon) \le 1$, so W_n is a nonnegative supermartingale with $W_0 = 1$ and Ville's inequality applies. The single attempted candidate is selected from screening-phase information alone, which is independent of the fresh certification probes, so the bound is unaffected by the selection. □

Remark 1 (Why exactly one attempt) Attempting k survivors in sequence at the shared threshold $1/\alpha$ would control the family-wise certification error only at level $k\alpha$ by a union bound — and the selection of the second attempt would depend on the first attempt's fresh probes, breaking the independence argument above. One attempt per run keeps the level exact and the argument honest; a multi-attempt variant is possible at per-attempt thresholds k/α (level α/k each), at proportionally higher probe cost. Our implementation enforces the single-attempt rule; an alarm during the attempt hard-falsifies the candidate (the checker is exact) and the run abstains from certification.

Remark 2 (What is and is not new here) Theorem 3 is elementary, and we prefer to say so. Its wealth process becomes permanently zero after a single alarm, so operationally it *is* the classical zero-failure binomial rule — "certify after n consecutive clean instances" — restated as an e-process. The e-process phrasing is what lets it compose with the rest of the framework: it may be stopped at any time, it inherits Ville's inequality without a fixed sample size, and, because its probes are drawn fresh after screening ends, the candidate it is applied to may be chosen by an arbitrarily adaptive screening procedure. That composition — an adaptive, budget-routed falsification stage feeding a selection-robust acceptance test — is the contribution; the acceptance test itself is a familiar object.

Theorem 3 is the answer to "what guarantee do I get on the model I finally accept": at $\alpha = \varepsilon = 0.05$, 59 consecutive fresh-oracle passes certify, at anytime level α, that the accepted model's disagreement rate with the specification on $\mathcal{D}_\mu$ is below 5%; at $\varepsilon = 0.1$, 29 passes suffice. Its scope is exactly its assumptions: an exact checker, and error measured on the enumerable distribution $\mathcal{D}_\mu$ in optimum-value terms — faithfulness on instances beyond enumeration reach is covered only through tiers B/C's falsification pressure, not by this certificate.

Proposition 4 (Pool FDR control) *Let $\{E(m)\}$ be the terminal e-values of the candidates of one screening run (a fixed pool). Applying e-BH at level α controls the false discovery rate of the rejected set at α, regardless of the dependence induced by shared probes, common instances, or pool entanglement. (The rejected set's FDR is the fraction of* faithful *models* *among the rejected; e-BH makes no statement about unfaithful models among the survivors — that is Theorem 3's job.)*

Proof Each $E(m)$ is a valid e-value for $H_0(m)$ by Theorem 2 (optional stopping preserves $\mathbb{E} \le 1$), *under the same conditioning*: if Assumption 1(c) fails, the tier-C factors need not be e-variables, the terminal $E(m)$ need not be an e-value, and no FDR statement survives — the caveat of Theorem 2 propagates verbatim. Given valid e-values, the e-BH procedure controls FDR for arbitrary dependence [9]. Extending the statement to adaptively grown pools (data-dependent numbers of hypotheses from repair rounds) requires online multiple-testing machinery we neither claim nor implement here; repair is future work. $\square$

Proposition 5 (Budget to rejection) *Fix an unfaithful candidate m and a routing rule under which the probes applied to m satisfy $\mathbb{E}[\log e_t \mid \mathcal{F}_{t-1}] \ge \Delta > 0$ and $|\log e_t| \le L$. Then the number τ of probes until rejection satisfies $\mathbb{E}[\tau] \le \big(\log(1/\alpha) + L\big)/\Delta$.*

Proof $M_t = \log E_t(m) - t\Delta$ is a submartingale minorant; optional stopping for the bounded-increment stopped process at $\tau \wedge n$ gives $\Delta\,\mathbb{E}[\tau \wedge n] \le \mathbb{E}[\log E_{\tau\wedge n}] \le \log(1/\alpha) + L$, and monotone convergence lets $n \to \infty$. $\square$

Proposition 5 is the quantitative content of "stop the moment evidence suffices": the budget scales as $\log(1/\alpha)$, so tightening the guarantee is cheap — consistent with the flat detection curves across α observed in Section 7.6.

Finally, a remark on the router's analytical status, stated with the honesty it requires. The deployed rule (2) is Kelly-*style*: a plug-in heuristic with a working posterior and calibrated working rates, not a provably growth-optimal policy, and Section 7.1 shows that at near-homogeneous probe costs the simpler cost-blind variant can match or beat it. What can be said rigorously concerns an online-learning relative under an explicit idealization.

Proposition 6 (No-regret tier selection, full-information idealization) *Fix a candidate. Suppose that after each step the realized log-growth per unit cost $r_t(p) = \log e_t(p)/c(p) \in [-L, L]$ is observable for* every *tier $p \in \{A, B, C\}$ (full-information feedback — e.g. when all tiers are probed in parallel for logging, or in simulation). Selecting tiers by Hedge [22, 23] with learning rate $\eta = \sqrt{8 \ln 3 / T}$ guarantees*

$$\max_p \sum_{t=1}^{T} r_t(p) \;-\; \mathbb{E}\Big[\sum_{t=1}^{T} r_t(p_t)\Big] \;\le\; L\sqrt{2T \ln 3},$$

and tier selection remains $\mathcal{F}_{t-1}$-measurable, so Theorem 2 is unaffected.

Proof Map $r_t \in [-L, L]$ to $\tilde r_t = (r_t + L)/(2L) \in [0, 1]$; the standard Hedge bound for 3 experts [23, Thm. 2.2] gives regret at most $\sqrt{(T/2)\ln 3}$ on the $\tilde r$ scale, and regret is positively homogeneous, so multiplying back by the range $2L$ yields $2L\sqrt{(T/2)\ln 3} = L\sqrt{2T \ln 3}$ on the original scale. Predictability holds because the weights at step t use only rewards observed before t. $\square$

In deployment only the selected tier's outcome is observed (bandit feedback), so Proposition 6 does not directly govern the running system; adversarial-bandit analogues with importance-weighted estimates would, at the usual $O(\sqrt{TK \log K})$ cost, and we leave that variant — and the question of whether it beats the plug-in heuristic in practice — open. Validity never depends on this choice.

Proposition 7 (Probe-family validity) *For the five problem families of Section 7, the implemented metamorphic relations — capacity/deadline/cost monotonicity, structural relaxation (conflict, edge, or coverage-row deletion), restriction (edge addition), and consistent data permutation — have null alarm probability exactly zero when both sides are solved exactly.*

Proof sketch Each relation is proved at specification level; exact solves transfer it to any faithful candidate. Monotone relaxations follow from feasible-set inclusion (e.g. enlarging a knapsack capacity or deleting a conflict pair only enlarges the feasible set of the specification, so the maximum cannot decrease); objective monotonicity follows from pointwise dominance; permutation invariance follows from the specification's symmetry under consistent relabeling of items, sets, vertices, or jobs. The one delicate case is deadline extension in on-time job selection, where the constraint *indices* shift with the deadline order: there, validity follows because the earliest-due-date feasibility conditions are equivalent to true single-machine schedulability of the selected set, and schedulability is monotone in deadlines. Full proofs accompany the probe library. $\square$

Remark 3 Assumption 1 for tier C deserves emphasis: the null alarm event "a wrong majority forms among partners" depends on the surviving pool, so its conditional probability is bounded uniformly only if calibration reflects the pool compositions actually encountered. Section 7.7 shows empirically that this is not pedantry — the conservative margin visibly erodes exactly, and only, when tier C is made the sole evidence source.

6 A verified probe library for five problem families

The guarantees of Section 5 are only as real as Assumption 1, and tier B's half of that assumption is a per-relation *proof obligation*. This section discharges it for a concrete library spanning five combinatorial families chosen to exercise packing, covering, colouring, assignment, and scheduling structure. For each family we implement a reference model, an independent executable checker (the tier-A enumeration oracle), and the metamorphic relations below; full validity proofs are in Appendix A. Table 2 summarizes.

Three structural remarks.

First, the relations fall into three provable classes — *feasible-set monotonicity* (relaxing a resource or deleting a structural restriction enlarges the specification's feasible set), *objective dominance* (a pointwise-increased cost vector cannot lower a minimum), and *symmetry* (consistent relabeling of indexed data is a bijection of the feasible set preserving objectives). These classes, not the individual relations, are what transfers

Table 2 The probe library. MRs state a provable relation between the specification optima of an instance I and its transform I'; $\uparrow$ ($\downarrow$) means the optimum cannot decrease (increase), $=$ means it is unchanged. Mutation classes are the functional corruptions used to create ground-truth unfaithful candidates.

Family (sense)	Metamorphic relations	Mutation classes
Knapsack with conflicts (max)	capacity$+\delta$: $\uparrow$; drop conflict: $\uparrow$; permute items: $=$	capacity shift; weight/value entry; vector rotation; conflict-row drop/half-drop; operator flip
Set cover (min)	drop element: $\downarrow$; cost entry$+\delta$: $\uparrow$; permute sets: $=$	cost entry; coverage-row drop/half-drop; operator flip
Graph colouring (min)	drop edge: $\downarrow$; add edge: $\uparrow$; relabel vertices: $=$	edge drop/half-drop; adjacency-logic flip
Generalized assignment (min)	capacity$+\delta$: $\downarrow$; cost entry$+\delta$: $\uparrow$; permute tasks: $=$	size/capacity entry; vector rotation; cost-matrix entry; operator flip
On-time job selection (max)	deadline$+\delta$: $\uparrow$; weight$+\delta$: $\uparrow$; processing$-\delta$: $\uparrow$; permute jobs: $=$	processing/deadline/weight entry; vector rotation; operator flip

to new families: any specification whose feasible set is downward-closed in a resource parameter inherits the first class verbatim, which is how the library should be extended toward global constraints such as `alldifferent` (invariant under value permutation) and `cumulative` (monotone in capacity and in deadline vectors).

Second, one relation is genuinely delicate and is the reason proofs are owed rather than waved at: deadline extension in on-time selection changes not only a right-hand side but the *index sets* of the earliest-due-date constraints, so monotonicity of the constraint system is not syntactically obvious; validity is restored by the classical equivalence between the EDD inequalities and true single-machine schedulability, which *is* monotone in deadlines (Appendix A). Metamorphic relations must be proved against the specification, not read off the formulation — this is precisely the failure mode that turns a would-be hard certificate into a silent source of false rejections.

Third, we quantify the manual investment the library represents, because the framework relocates rather than eliminates part of the oracle problem: each family required a reference builder, an independent checker, instance generators, and three-to-four proved relations — roughly 150–250 lines of code and proofs of a few lines each, a person-day per family for someone fluent in the problem class. This is the honest price of tiers A and B; it is paid once per problem *family* and amortizes over every candidate, every instance, and every certification ever run against it.

Fourth, the mutation classes mirror the errors observed from real LLM modellers (Section **??**) at the semantic rather than syntactic level: misread constants, positionally corrupted data, misaligned columns (rotation), dropped constraint structure, and inverted comparisons. Each corruption is *instance-generic* — applied identically to every instance the candidate ever sees — so a mutant is a wrong *model*, not a wrong instance, exactly like a systematic LLM misreading.

7 Experimental evaluation

Implementation.

All experiments run on CPMpy [10] with the OR-Tools CP-SAT solver [11]. Five problem families are implemented, each with a reference model builder, an *independent* pure-Python specification checker (the tier-A enumeration oracle), micro/mid instance generators, and the family's metamorphic relations of Proposition 7: 0/1 knapsack with pairwise conflicts, minimum-cost set cover, graph colouring (minimizing the number of colours), generalized assignment, and weighted on-time job selection on a single machine. Probe costs are measured in *probe-seconds*: the wall-clock time of all computation a probe consumes — CP-SAT solves and Python brute-force enumeration alike (we avoid "solver-seconds," which would suggest solver calls only). Code, calibration corpora, all raw results, and the cached LLM generations are provided as supplementary material. An evaluation instrument, separate from the protocol, supports Section 7.1: `estimate_error_rate` draws 400 fresh micro instances for a given candidate, scores each against the enumeration oracle, and returns the empirical disagreement rate with a Wilson interval. It is never consulted by the certifier — it exists so that claims about certified models are measured rather than inferred from construction labels. Two further artifacts ship with the code: a pinned environment (the reported numbers come from Python 3.9.6, CPMpy 0.9.29, OR-Tools 9.14.6206 on arm64 macOS, and the result files record that provenance together with the git commit), and `check_artifacts.py`, which fails loudly if the compiled PDF is older than its sources, if the shipped results are a smoke run rather than a publication run, if the run came from a dirty working tree, if Table 4's per-policy figures disagree with the results file, or if a quoted interval fails to recompute from the audit log's counts at the recorded z. Each check asserts a minimum number of matches, so a check that finds nothing fails rather than passing silently. Runs are published transactionally: all outputs are written to a staging directory, checked for completeness, hashed into a manifest, and only then promoted, so the canonical package can never mix calibration from one run with results from another. One reproducibility caveat: the per-pool budget is set to 80 times the *measured* mean probe cost, so it drifts by a few percent between machines and runs (263–286 ms across our single-threaded reruns; it was larger before the solver thread count was pinned) and downstream counts move by a candidate or two; the qualitative findings and every interval reported here are stable across our reruns, and a fixed probe-count budget removes the drift for anyone needing bit-exact replication.

Unfaithful candidates by functional mutation.

To measure error rates one needs ground truth, so the first two studies replace the LLM with *functional corruptions* of the reference builder — systematic, instance-generic errors of the kinds LLM modellers actually make: a bound misread by a constant, one vector entry corrupted positionally, a data-wiring misalignment (vector rotation), a dropped structural row or half-family, a corrupted cost-matrix entry, and a

Table 3 Calibration corpus (two held-out families, 240 trials per tier and hypothesis). $\hat{p}_0/\hat{p}_1$ are observed alarm rates on faithful/mutant candidates; $\bar{p}_0$ is the Clopper–Pearson 97.5% upper bound the bets use; p_1 is the *working* detection rate the bets wager with, deliberately shrunk, $p_1 = \max\{0.8\,\hat{p}_1,\ \min(3\bar{p}_0, 0.5),\ 0.15\}$ — misspecifying p_1 affects power only, never validity (Lemma 1); cost is measured mean probe time.

Tier	$\hat{p}_0$ (faithful)	$\bar{p}_0$	$\hat{p}_1$ (mutants)	p_1 (bet)	mean cost
A (micro-oracle)	0/240	0.0153	0.592	0.473	3.1 ms
B (MRs)	0/240	—	0.042	—	2.6 ms
C (cross-majority)	28/240	0.164	0.567	0.493	5.0 ms

flipped comparison operator in one constraint family. Mutants are screened for semantic difference from the reference (equivalent mutants are resampled), the standard mutation-testing practice.

Calibration, and what α means in the tables below.

Betting parameters are calibrated on two held-out families (knapsack with conflicts, graph colouring) and evaluated on the three *disjoint* remaining families — so every result below also tests whether conservative calibration transfers across problem families. Throughout the experiments we write $\alpha = 0.05$; recall from Section 5 that for runs in which the pool-entangled tier C participates this level is *conditional* on Assumption 1(c) — which these experiments stress-test rather than establish — whereas the certification guarantee of Theorem 3, which uses the exact tier-A oracle alone, is unconditional at α. Table 3 reports the measured corpus. Two entries deserve comment. Tier B's 0/240 row is not an estimate but a *consistency check* of Proposition 7 — a single faithful alarm would indicate an invalid relation and aborts calibration. Tier C's null alarm rate of 11.7% is the empirical face of pool entanglement: even a faithful candidate is outvoted when two of its three sampled partners happen to agree on a wrong value; the Clopper–Pearson bound $\bar{p}_0 = 0.164$ is what the bets then use.

7.1 Mutation testbed: validity, power, and routing

Pools of $k = 6$ candidates (each faithful or a screened mutant with probability 1/2) are screened at $\alpha = 0.05$ under a shared screening allowance of 286 ms of measured probe time per pool, then the best survivor — and only that one candidate, per Remark 1 — enters ε-certification ($\varepsilon = 0.10$, up to 70 tier-A-cost units). The allowance is a *launch threshold* rather than a hard cap: a probe is started while allowance remains and its measured cost is charged afterwards, so a run may overshoot by at most one probe — and since a single tier-A or tier-B probe can be expensive, that one probe matters. The result files record realized expenditure: mean 286–287 ms against the 286 ms allowance, with a maximum of 423 ms over the 300 runs. We report the allowance and the realized spend rather than pretending the former bounds the latter. Pools are generated from a per-replication seed and are therefore *identical* across policies; probe randomness is per (policy, replication). 100 replications per policy; Table 4 reports the results with Wilson 95% intervals on the headline rates.

Table 4 Mutation testbed (100 replications/policy, $\alpha = 0.05$, $\varepsilon = 0.10$, real CP-SAT probes; calibration and evaluation families disjoint; identical pools across policies). False rejection is over faithful candidates ($n = 293$/policy); detection over mutants ($n = 307$), counting screening and certification-phase falsifications. Kill cost appears twice: screening probes only, and including certification probes spent on a mutant that survived screening. The last three columns classify every issued certificate by an *independent* audit of the certified model's error rate (Section 7.1), not by its construction label.

Routing	False rej. [CI]	Detection [CI]	Kill cost		Certif.	audit vs. ε		
			screen	+cert	/100	below	above	inconcl.
Kelly	0.000 [0, .013]	0.958 [.929, .975]	15	15 ms	96	96	0	0
round-robin	0.000 [0, .013]	0.870 [.827, .903]	16	16 ms	99	99	0	0
cost-blind	0.000 [0, .013]	0.954 [.925, .973]	14	15 ms	94	94	0	0

Validity holds without an observed exception — none of 879 faithful-candidate exposures across the three policies was rejected (a finite-sample zero, hence the intervals) — despite the bets being calibrated on different problem families than those evaluated. Detection is 89–96% within a sub-second screening budget.

Auditing the certificates.

A certificate is not proof that the certified model satisfies $\mathrm{err}(m) < \varepsilon$: certifying a candidate with $\mathrm{err}(m) \geq \varepsilon$ is exactly the type-I event Theorem 3 bounds by α. We therefore *measure* rather than infer. Every issued certificate triggers an independent audit — 400 fresh micro instances from $\mathcal{D}_\mu$, scored against the enumeration oracle, with a Wilson interval (Section 7, `estimate_error_rate`). *All* audits use the two-sided 97.5% level ($z = 2.2414$), not only the two highlighted below: the wider interval is the conservative choice for the classification that matters here, since declaring an interval to lie entirely below ε becomes harder, never easier. Across the 300 runs, 289 ended in a certificate and 11 abstained. Every one of those 289 audits places its interval *entirely below* $\varepsilon = 0.10$: none lies above, and none is inconclusive at this sample size. In this run no certified model was a mutant — the acceptance stage's fresh oracle probes either falsified the attempted survivor or the survivor was faithful — so there is no near-miss case to display, and we quote no per-candidate interval the audit log does not contain.

That outcome is cleaner than the guarantee requires, and it would be a mistake to read it as the guarantee's content. Theorem 3 promises a *tolerance*, not identity: a candidate disagreeing with the specification on fewer than an ε-fraction of micro instances may certify legitimately, and a candidate with $\mathrm{err}(m) \geq \varepsilon$ may certify with probability at most α per run. Both are events this testbed can produce: an earlier run of the same code under a different solver configuration (multi-threaded, hence a larger derived allowance) certified two near-miss mutants, audited at $\widehat{\mathrm{err}} = 0.040$ and 0.108. We mention that to characterize the tolerance and deliberately do not import its numbers into the tables, which report one configuration end to end. Readers who want a tighter promise buy it with probes: $\lceil \log(1/\alpha)/\log \frac{1}{1-\varepsilon} \rceil$ passes per attempt, i.e. 59 at $\varepsilon = 0.05$ and 299 at $\varepsilon = 0.01$.

Cost of the acceptance stage.

Certification is deterministic in length: a successful attempt consumes exactly $\lceil \log(1/\alpha)/\log \frac{1}{1-\varepsilon} \rceil = 29$ fresh tier-A probes, i.e. $\approx$91 ms at this run's measured mean tier-A cost of 3.15 ms, about a third of the 286 ms screening allowance on top of it; a failed attempt costs less, stopping at the first alarm. Certification probes barely move the kill-cost column because few mutants reach the acceptance stage at all.

On routing: at this testbed's probe economics the measured tier costs span only a factor 2.4, and cost-blind selection matches Kelly while plain round-robin trails by seven detection points — cost-normalization is not what separates policies here; Section 7.11 isolates the routing question in a controlled heterogeneous-cost regime, and we flag plainly that the Kelly rule's advantage is regime-dependent.

7.2 Two different questions: procedure vs. outputs

The audit above and the guarantee of Theorem 3 answer different questions, and conflating them would be a category error.

Procedural validation asks whether the *rule* has the advertised error rate: run the certification procedure repeatedly against candidates whose error is known to sit at or above ε, and measure how often a certificate is nonetheless issued. Theorem 3 says this frequency is at most α. The implementation's test suite contains a Monte-Carlo *sanity check* of this — tier-A outcomes mocked as Bernoulli draws at exactly the boundary $\mathrm{err} = \varepsilon$, four seeds $\times$ 250 simulated runs — with realized certification frequencies 0.064, 0.036, 0.044, 0.044 and pooled 0.047 over 1000 runs. We describe it as a sanity check and nothing more: at $n = 1000$ the Monte-Carlo standard error at α is $\approx$0.007, so this experiment cannot resolve 0.05 from, say, 0.06, and one seed exceeds α as sampling noise alone would predict. The inequality is established by Theorem 3; the simulation only guards against an implementation that violates it grossly.

Output auditing asks a question about the particular models this system actually accepted: does *this* certified candidate satisfy $\mathrm{err}(m) < \varepsilon$? That is what Section 7.1 measures on fresh instances. A clean audit does not prove the procedure valid (it could be luck), and a straddling interval does not prove it invalid (it is the type-I event the procedure is allowed). Reporting both, separately, is the honest presentation: the theorem governs the rule, the audit inspects the artifacts.

7.3 Validity across the operating range

Sweeping $\alpha \in \{0.01, 0.02, 0.05, 0.10, 0.20\}$ (100 replications per level, Kelly routing) yields empirical false-rejection 0.0000 at *every* level ($n = 280$–317 faithful candidates per level), with detection between 87% and 96% and no trend against α beyond sampling noise; Figure 5 plots both curves. Two remarks.

First, the guarantee of Theorem 2 is an upper bound, not a calibration claim: ECLAIR promises *at most* α, and the observed margin is the structural price of a composite null — tier B cannot false-alarm by proof, and tiers A/C bet against upper-bounded null rates because the true rates are unknowable. Second, the flatness of the detection curve is Proposition 5 in action: rejection budgets grow only logarithmically as α tightens.

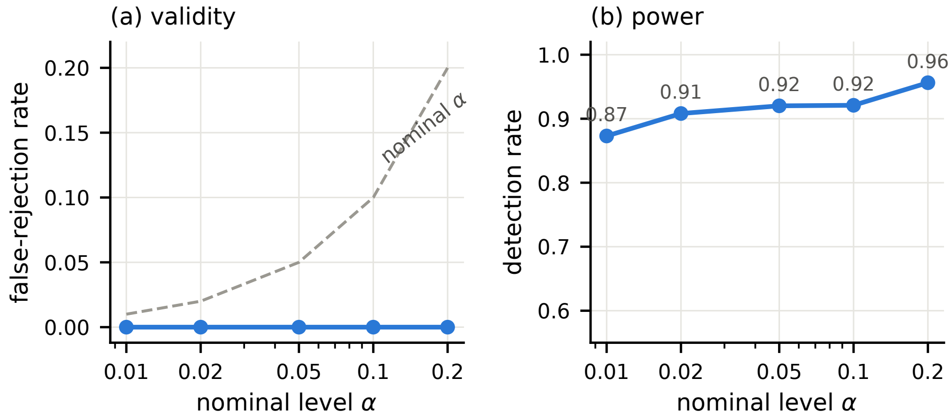

Fig. 3 Validity and power across the operating range (100 replications per level, Kelly routing). (a) Empirical false-rejection of faithful candidates sits at zero under the nominal-α diagonal at every level — the guarantee is an upper bound and holds with the conservative margin the composite null demands. (b) Detection of mutants is flat in α, the empirical face of the logarithmic budget bound of Proposition 5

Table 5 Probe-tier ablation (100 replications/configuration, Kelly, $\alpha = 0.05$).

Tiers	False rej.	Detection	Kill cost
A+B+C	0.0000	0.884	14 ms
A+B	0.0000	0.904	15 ms
A+C	0.0000	0.853	14 ms
B+C	0.0000	0.675	15 ms
A	0.0000	0.876	14 ms
C	0.0399	0.625	32 ms

7.4 Tier ablation: what each probe class buys

Table 6 repeats the testbed with restricted probe pools; Figure 6 visualizes the two headline columns.

Three findings. *(i) Entanglement consumes the conservative margin.* Every configuration containing an unentangled tier records zero false rejections; tier C alone records 0.0401 — still within $\alpha = 0.05$, but with the margin that conservative calibration provides everywhere else fully spent. This is the empirical face of the Remark after Proposition 7. We are careful about what it does *not* say: a valid tier does not immunize an invalid one — multiplying sound A/B factors by unsound C factors invalidates the product — so tier C is valid, alone or in combination, *only* conditionally on Assumption 1(c). What the ablation shows is narrower and still useful: empirical false rejections appear at the margin precisely when cross-model evidence carries the process alone, which is where the assumption is most strained and where

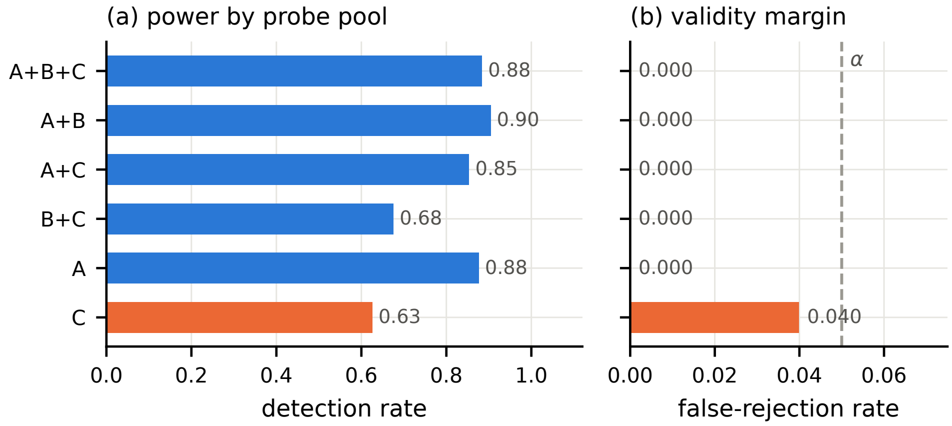

Fig. 4 Probe-tier ablation (100 replications per configuration, Kelly, $\alpha = 0.05$). (a) Detection by probe pool. (b) The validity margin: every pool containing an unentangled tier (blue) records zero false rejections; the pool-entangled cross-model tier alone (orange) stays within α but consumes the entire conservative margin. Every configuration including tier C is valid only *conditionally* on Assumption 1(c); the figure reports empirical behaviour, not evidence for that assumption

pool-composition-conditioned calibration should be required first. *(ii) The exact oracle carries most of the power, and the metamorphic tier adds to it.* Dropping tier B (A+C at 0.853 vs. A+B+C at 0.884) costs three points of detection although tier-B alarms fire on only $\sim$4% of mutant probes — a rare hard certificate is worth more than its rate suggests, because it needs no accumulation. Dropping tier A is far more damaging (B+C at 0.675): the enumeration oracle is the workhorse. *(iii) Cheap cross-model evidence can cost power, not just validity.* A+B (0.904) slightly *exceeds* the full pool (0.884), and tier A alone (0.876) is within a point of it, so the router's willingness to spend on tier C is not free: cross-model probes are the noisiest evidence per probe-second here, and including them draws budget away from the oracle. Combined with the validity picture above — tier C is the only configuration whose false rejections reach the margin — the practical recommendation is narrow: use cross-model checks when no exact oracle is available for a family, not as a supplement when one is. (In the earlier multi-threaded configuration the ordering favoured the full pool; the comparison is sensitive to relative probe costs, which is precisely why the thread count is pinned and reported.)

7.5 Two different questions: procedure vs. outputs

The audit above and the guarantee of Theorem 3 answer different questions, and conflating them would be a category error.

Procedural validation asks whether the *rule* has the advertised error rate: run the certification procedure repeatedly against candidates whose error is known to sit at or above ε, and measure how often a certificate is nonetheless issued. Theorem 3

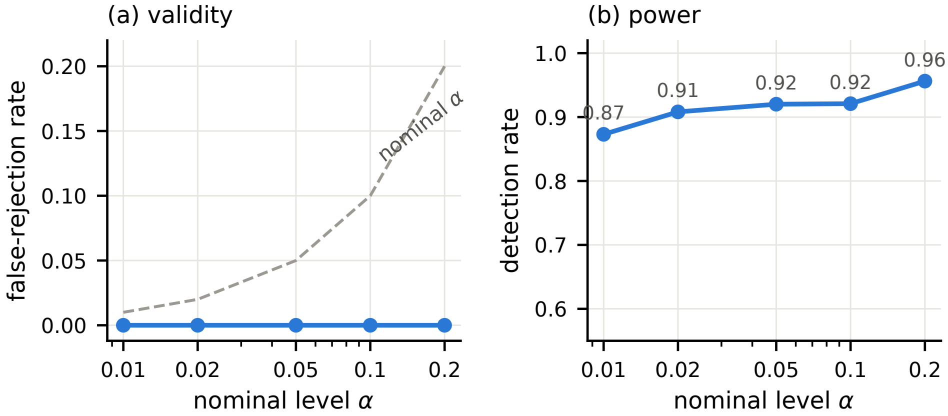

Fig. 5 Validity and power across the operating range (100 replications per level, Kelly routing). (a) Empirical false-rejection of faithful candidates sits at zero under the nominal-α diagonal at every level — the guarantee is an upper bound and holds with the conservative margin the composite null demands. (b) Detection of mutants is flat in α, the empirical face of the logarithmic budget bound of Proposition 5

says this frequency is at most α. The implementation's test suite contains a Monte-Carlo *sanity check* of this — tier-A outcomes mocked as Bernoulli draws at exactly the boundary $\mathrm{err} = \varepsilon$, four seeds $\times$ 250 simulated runs — with realized certification frequencies 0.064, 0.036, 0.044, 0.044 and pooled 0.047 over 1000 runs. We describe it as a sanity check and nothing more: at $n = 1000$ the Monte-Carlo standard error at α is $\approx$0.007, so this experiment cannot resolve 0.05 from, say, 0.06, and one seed exceeds α as sampling noise alone would predict. The inequality is established by Theorem 3; the simulation only guards against an implementation that violates it grossly.

Output auditing asks a question about the particular models this system actually accepted: does *this* certified candidate satisfy $\mathrm{err}(m) < \varepsilon$? That is what Section 7.1 measures on fresh instances. A clean audit does not prove the procedure valid (it could be luck), and a straddling interval does not prove it invalid (it is the type-I event the procedure is allowed). Reporting both, separately, is the honest presentation: the theorem governs the rule, the audit inspects the artifacts.

7.6 Validity across the operating range

Sweeping $\alpha \in \{0.01, 0.02, 0.05, 0.10, 0.20\}$ (100 replications per level, Kelly routing) yields empirical false-rejection 0.0000 at *every* level (n = 280–317 faithful candidates per level), with detection between 87% and 96% and no trend against α beyond sampling noise; Figure 5 plots both curves. Two remarks.

First, the guarantee of Theorem 2 is an upper bound, not a calibration claim: ECLAIR promises *at most* α, and the observed margin is the structural price of a composite null — tier B cannot false-alarm by proof, and tiers A/C bet against upper-bounded null rates because the true rates are unknowable. Second, the flatness of the

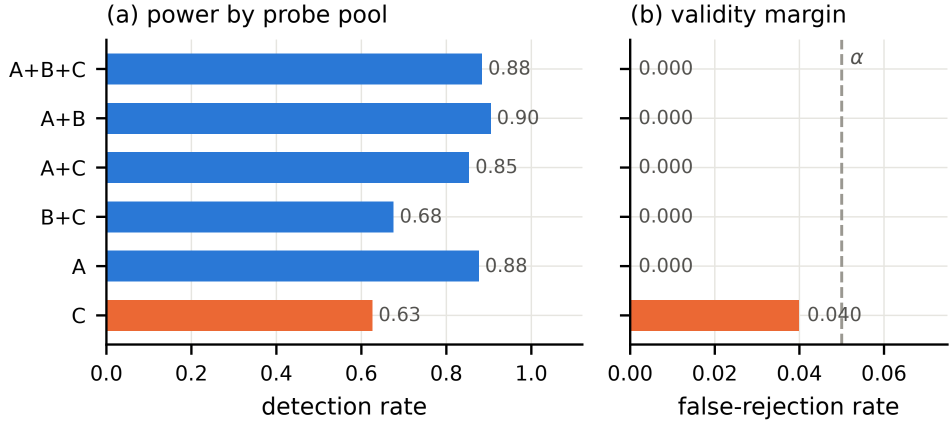

Fig. 6 Probe-tier ablation (100 replications per configuration, Kelly, $\alpha = 0.05$). (a) Detection by probe pool. (b) The validity margin: every pool containing an unentangled tier (blue) records zero false rejections; the pool-entangled cross-model tier alone (orange) stays within α but consumes the entire conservative margin. Every configuration including tier C is valid only *conditionally* on Assumption 1(c); the figure reports empirical behaviour, not evidence for that assumption

Table 6 Probe-tier ablation (100 replications/configuration, Kelly, $\alpha = 0.05$).

Tiers	False rej.	Detection	Kill cost
A+B+C	0.0000	0.884	14 ms
A+B	0.0000	0.904	15 ms
A+C	0.0000	0.853	14 ms
B+C	0.0000	0.675	15 ms
A	0.0000	0.876	14 ms
C	0.0399	0.625	32 ms

detection curve is Proposition 5 in action: rejection budgets grow only logarithmically as α tightens.

7.7 Tier ablation: what each probe class buys

Table 6 repeats the testbed with restricted probe pools; Figure 6 visualizes the two headline columns.

Three findings. *(i) Entanglement consumes the conservative margin.* Every configuration containing an unentangled tier records zero false rejections; tier C alone records 0.0401 — still within $\alpha = 0.05$, but with the margin that conservative calibration provides everywhere else fully spent. This is the empirical face of the Remark after Proposition 7. We are careful about what it does *not* say: a valid tier does not immunize an invalid one — multiplying sound A/B factors by unsound C factors invalidates the product — so tier C is valid, alone or in combination, *only* conditionally on

Table 7 Baselines under the fair design (300 replications, $\alpha = 0.05$, identical pools, Wilson 95% intervals). Groups share a probe diet; comparisons across groups are budget-matched but not probe-matched.

Method	False rej. [CI]	Detection [CI]	Pick acc.
ECLAIR (all tiers)	0.000 [.000, .004]	0.933 [.915, .948]	0.936
seq. (same stream)	0.000 [.000, .004]	0.863 [.840, .884]	—
fixed-n (same stream)	0.000 [.000, .004]	0.840 [.815, .862]	—
ECLAIR-C	0.049 [.037, .066]	0.616 [.584, .647]	0.742
fixed-sample C	0.078 [.062, .098]	0.602 [.570, .633]	0.651
peeking C	0.087 [.070, .108]	0.620 [.589, .651]	—
split conformal C	0.013 [.007, .023]	0.318 [.289, .349]	—

Assumption 1(c). What the ablation shows is narrower and still useful: empirical false rejections appear at the margin precisely when cross-model evidence carries the process alone, which is where the assumption is most strained and where pool-composition-conditioned calibration should be required first. *(ii) Hard certificates matter beyond their rate.* Removing tier B (A+C vs. A+B+C) costs about ten points of detection although tier-B alarms occur on only $\sim$4% of mutant probes. *(iii) A caveat on tier A's dominance.* Tier A alone detects 95% here, but mutation screening guarantees every mutant is micro-instance-detectable — precisely tier A's regime. Errors of LLM origin need not be (conventions that diverge only at scale; see below), which is why the tiered design is retained.

7.8 External baselines at matched budget

The routing policies of Table 4 are internal ablations; this section compares against the strategies a practitioner would actually reach for, under a design built for fairness: every method sees the *identical* candidate pools (pool seeds depend only on the replication index), comparisons are grouped by probe diet so that no method is judged against evidence it was not given, and every rate carries a Wilson 95% interval (300 replications; $n = 870$ faithful and 930 mutant exposures per method). Three comparison groups.

Group 1 — the accounting, isolated. Each candidate receives one fixed round-robin stream of 13 probes across tiers A, B, C with e-factors recorded; the *same stream* is then scored two ways: sequentially (reject iff the running product ever reaches $1/\alpha$; Ville-valid) and at fixed sample size (reject iff the endpoint product reaches $1/\alpha$; Markov-valid). Identical probes, identical e-factors, identical threshold; the sequential rejection set contains the fixed-sample one by construction.

Group 2 — tier-C-only methods, like for like. The framework restricted to cross-model probes (ECLAIR-C) against the binomial fixed-sample test at the exact critical value under $\bar{p}_0^C$, its uncorrected-peeking variant (the same test applied after every probe), and split conformal against a 200-run faithful calibration sample.

Group 3 — no-rejection strategies. Single-shot acceptance and self-consistency majority voting over eight shared instances.

The groups support three precise statements.

First, with probes and e-factors held literally identical, sequential accounting dominates fixed-sample accounting for free. Both sit at zero false rejections; the sequential rejection set contains the fixed-sample one by construction (detection 0.863 vs. 0.840, i.e. 22 additional mutants of 930); and the sequential rule *would stop* after 4.4 probes per mutant on average against the fixed test's mandatory 13 — the same evidence at one third of the budget. (Both arms execute all 13 probes here so the comparison is exactly paired; the 4.4 is the crossing time recorded on that shared stream, i.e. what a properly stopped implementation would consume, not what this experiment executed.)

Second, on the pool-entangled tier the fixed-schedule tests overrun their nominal level, and at 300 *replications we can say so with confidence.* The binomial fixed-sample test realizes 0.078 with a Wilson interval $[.062, .098]$ lying entirely above $\alpha = 0.05$, and its uncorrected-peeking variant 0.087 $[.070, .108]$, also entirely above; the e-process restricted to the same tier sits at 0.049 $[.037, .066]$, straddling α. We are careful about the mechanism, because it is *not* a claim that sequential testing is magic: all four methods bet against the same calibrated bound $\bar{p}_0^C$, which was estimated on different problem families, and Assumption 1(c) is exactly what transfer can break. A hard accept/reject threshold spends its whole margin the moment the bound is optimistic, whereas a supermartingale carries conservative slack (Lemma 1) and degrades more gracefully. The honest reading is that this experiment measures how the two accountings respond to a mis-specified null, and that the e-process response is the safer one — not that the e-process is valid where its assumption fails.

Third, the no-rejection strategies illustrate what is being purchased. Majority voting picks a faithful model 81% of the time and silently hands over an unfaithful one the remaining 19%, with no level to quote and nothing resembling Theorem 3's certificate; single-shot is the coin flip its 0.461 suggests.

7.9 Robustness: families, transfer direction, pool composition

Three sweeps probe whether the headline numbers are artifacts of a particular experimental slice (100 replications each, Kelly, $\alpha = 0.05$; Table 8). *Per family*: validity records no false rejection in any evaluation family separately (finite-sample zeros); detection varies with family difficulty (0.98 for set cover and 0.96 for on-time selection against 0.79 for generalized assignment, whose cost-matrix mutations produce the subtlest optimum shifts) — power is problem-dependent, the guarantee is not. *Reversed calibration split*: recalibrating on the three former evaluation families and evaluating on knapsack+colouring yields false rejection 0.0000 with detection 0.829 ($n = 291$ faithful) — cross-family transfer of the conservative bounds is bidirectional, not an artifact of which families happened to calibrate. *Pool size and routing prior*: across $k \in \{4, 6, 10\}$ and routing priors $q_{\text{prior}} \in \{0.3, 0.5, 0.7\}$, false rejection stays at 0.0000 while detection moves only through the router's aggressiveness (a pessimistic prior under-probes, 0.74; a suspicious one probes harder, 0.93) — the router's belief state touches power alone, never the level, exactly as Theorem 2 promises for any predictable routing.

Table 8 Robustness sweeps (100 replications per row, Kelly, $\alpha = 0.05$).

Sweep	Configuration	False rej.	Detection
family	set cover	0.0000	0.980
family	generalized assignment	0.0000	0.794
family	on-time selection	0.0000	0.958
transfer	reversed calibration split	0.0000	0.829
pool	$k = 4$	0.0000	0.934
pool	$k = 10$	0.0000	0.883
prior	$q_{\text{prior}} = 0.3$	0.0000	0.739
prior	$q_{\text{prior}} = 0.7$	0.0000	0.927

7.10 Probing the silent-error regime: ambiguous specifications

The main LLM study's lesson — loud failures, faithful survivors — raises the question of whether *silent* misreadings can be elicited at all. We rephrased three specifications to plant boundary ambiguities, fixed one canonical reading in the reference checker, and regenerated candidate pools (both model families, three prompt styles each): colouring with the objective convention sentence *removed* ("minimize the number of colors used"), knapsack with "the selected items together must be lighter than the capacity" (strict vs. non-strict), and on-time selection with "every selected job is finished before its deadline" (strict vs. non-strict completion).

The outcome is a negative result we report with its explanation: 10/18 generations passed intake, and *all ten* implemented the canonical reading — no silent other-reading model appeared. The framework behaved correctly throughout: 0/10 false rejections, the canonical reading picked in 3/3 families, and an ε-certificate issued for the canonical model in all three — the appropriate outcome when every candidate is in fact canonical, though (as in the main LLM study) an all-null pool tests correctness on nulls rather than detection. Two mechanisms account for the miss. First, one planted ambiguity is value-equivalent at the optimum: a model minimizing the *count* of distinct colours and one minimizing 1+max index return the same optimal value on every instance, by precisely the renumbering argument of Appendix A — the readings differ only off-optimum, where no value-level probe looks. Second, for the strict / non-strict boundary pairs, current models normalize aggressively to the conventional reading ($\leq$): the training distribution of textbook formulations evidently dominates the literal text. The constructive lesson for the elicitation agenda (Section 10) is that silent LLM errors are unlikely to hide in boundary conventions; structural underspecification — omitted side constraints, ambiguous aggregation, units — is the more promising hunting ground, and value-equivalent ambiguities require solution-level (not optimum-level) probes, which the framework accommodates as additional tier-A checkers.

7.11 Routing under heterogeneous probe costs

The CP-SAT testbed's probes all cost single-digit milliseconds, which compresses the routing question. To isolate it, a controlled simulation reproduces the certification layer with the measured alarm-rate structure but a 20:1 probe-cost spread (expensive

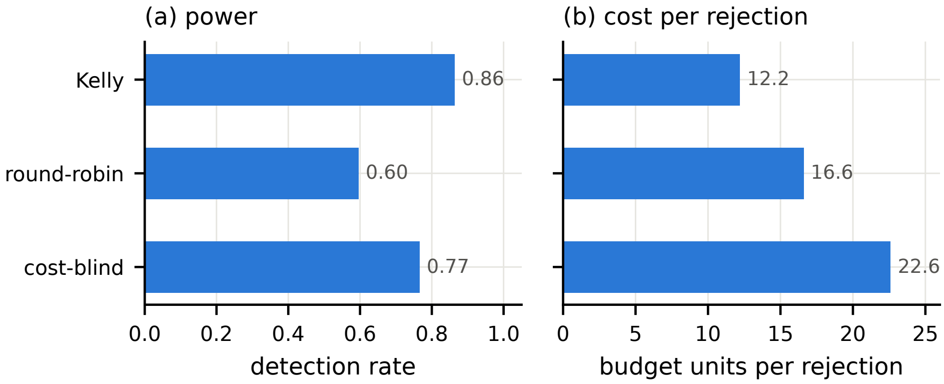

Fig. 7 Routing under a 20:1 probe-cost spread (500 replications, $\alpha = 0.05$; measured alarm-rate structure, controlled costs). Kelly routing dominates on both detection (a) and budget per rejection (b); false-rejection stays below α under every policy, as validity is routing-invariant

near-exact oracles vs. cheap noisy checks — the regime of industrial instances, of costly solver calls, and of LLM-as-judge probes priced per token). Over 500 replications (Figure 7): Kelly routing detects 86% of unfaithful candidates at mean cost 12.2 budget units per kill, against 60% at 16.6 for round-robin and 77% at 22.6 for cost-blind selection — while false rejection stays at 0.000–0.003, below α for every policy. Routing gains scale with probe-cost heterogeneity; validity is routing-invariant, as Theorem 2 requires.

8 Deployment guidance

Because every quantity in the framework is operational, the knobs a deploying team must set reduce to four, each with a concrete rule.

The certification tolerance ε.

The acceptance guarantee costs $\lceil \log(1/\alpha)/\log \frac{1}{1-\varepsilon} \rceil$ consecutive fresh-oracle passes per attempt (29 at ε=0.10, 59 at 0.05, 299 at 0.01, all at $\alpha = 0.05$), so ε is the knob trading oracle budget for the strength of the accepted model's certificate; it should be set from the application's tolerance for rare optimum-value disagreement on small instances. Note what the tolerance permits rather than what one run happened to show: a candidate disagreeing on fewer than an ε-fraction of micro instances may certify legitimately, so at loose ε a certified model is a *nearly*-agreeing model, not a provably identical one.

The level α.

By Proposition 5 and Figure 5, tightening α costs only logarithmic budget and essentially no detection — so α should be chosen by the downstream stakes. Note that α governs *both* sides: on the screening side it bounds wrongly discarded faithful models, and on the acceptance side it bounds the probability of certifying a candidate whose error rate is at least ε. The two acceptance knobs work together — ε fixes the tolerated disagreement rate, α the risk of exceeding it — and neither should be chosen from compute anxiety. We used 0.05 throughout; 0.01 behaved identically.

The calibration corpus.

Validity consumes exactly one number per noisy tier: an upper confidence bound $\bar{p}_0$ on the null alarm rate. With zero observed false alarms in n calibration trials, the Clopper–Pearson bound scales as $\bar{p}_0 \approx 3.7/n$ (n=100: 0.036; n=240: 0.015; n=1000: 0.0037), and a smaller $\bar{p}_0$ buys a larger alarm multiplier $p_1/\bar{p}_0$ — calibration effort converts directly into per-alarm evidence. Known-faithful models for the corpus are the cheap part: hand-written reference models on any families adjacent to the deployment domain suffice, and our results show the bounds transfer across families when used conservatively.

The budget.

The testbed's operating point — ∼80 mean-cost probes per pool of six — put the marginal probe well past the detection plateau. The budget-to-rejection bound gives the back-of-envelope sizing rule: expect $\sim \log(1/\alpha)/\Delta$ probes per unfaithful candidate, with Δ the per-probe expected log-evidence (≈1.5 for our calibrated tier A under $\hat{p}_1 \approx 0.6$), so single-digit probes per rejection — consistent with the measured 20 ms kill costs against 5 ms probes.

What to log.

The evidence ledger — every probe, its tier, instance seed, outcome, and e-factor — is the audit object: it makes each rejection reproducible (replay the counterexample), each certification quantifiable (the fresh-oracle wealth that crossed $1/\alpha$, and the ε it certifies), and each abstention actionable (which readings died on which probes). Teams wrapping an existing agentic modelling loop [3, 13] need to intercept only two things: candidate model source and solver calls.

9 Threats to validity

Internal. The screening guarantee rests on Assumption 1, whose parts (a)/(b) are proved in our instantiation but whose tier-C part is a deployment assumption; the ablation and the fair baselines localize exactly where it is fragile — with cross-model probes carrying sole weight, the tier-C-only e-process realizes 0.049 $[.037, .066]$ at $\alpha = 0.05$: inside the level, but with the conservative margin fully spent and the interval straddling α. Bucket-conditioned calibration of $\bar{p}_0^C$ (by pool composition) is the concrete remedy before tier C carries certification-critical weight. Tier-B validity is by proof, but only for relations proved against the *specification*; a relation read off a formulation can be invalid (Section 6). The certification theorem's scope is its assumptions: an exact checker and the enumerable instance distribution — Theorem 3 certifies optimum-value agreement on $\mathcal{D}_\mu$, nothing more.

Construct. Mutation screening makes every testbed fault micro-detectable, biasing tier importance toward A (Section 7.7); proxy faithfulness labels in the LLM studies are themselves enumeration-based and share this bias. The ambiguity study is the counterweight, since other-reading models differ precisely where enumeration on canonical micro-instances says they do.

External. Two model families, five problem families, one modelling language: claims about "LLM modellers" beyond these are extrapolation. The error taxonomy (Table **??**) will age as models improve; the framework's premise — that whatever errors remain be caught with a guarantee — does not depend on the taxonomy's contents.

10 Limitations and outlook

Calibration scope. Validity rests on Assumption 1; our evidence that conservative calibration transfers across problem families is empirical (zero false rejections under family transfer), not a theorem, and tier-C calibration should be conditioned on pool composition before cross-model evidence carries substantial weight — the ablation shows the margin, though never the bound, giving way. *Screened mutants.* The testbed's faults are guaranteed micro-detectable by construction; LLM-native silent errors need not be, and eliciting them requires deliberately ambiguous specifications — the planned abstention study, which also connects certification to conversational elicitation. *Checker provenance.* Our enumeration checkers are hand-written and independent; a fully LLM-generated pipeline must calibrate checker error as tier-A noise, which the framework accommodates but this study does not exercise. *Breadth.* Five families suffice to exercise every mechanism; scaling the probe library toward CSPLib/NL4Opt breadth, and stating the MR validity results for global constraints (`alldifferent`, `cumulative`) in the generality they deserve, are the natural next steps. *Beyond one modelling language.* Nothing in the framework is CPMpy-specific: candidates are opaque functions from instances to optima, so MiniZinc, MILP, or SMT candidates certify identically — indeed a heterogeneous pool (the same specification modelled in CP *and* MILP) strengthens tier C precisely because formalism diversity decorrelates errors. What each new language requires is an exact solve at micro/mid scale and, for tier B, relations proved against the specification; the three relation classes of Section 6 are formalism-independent. *Eliciting silent errors.* The ambiguity study (Section 7.10) shows boundary conventions are normalized away by current models and value-equivalent readings are invisible to optimum-level probes; the elicitation agenda should target structural underspecification, and the probe pool should grow solution-level checkers where readings coincide in value but not in argmax.

11 Conclusion

ECLAIR turns the verification of LLM-generated constraint models from a heuristic detection exercise into sequential hypothesis testing with explicit guarantees in both directions, kept carefully apart: anytime-valid control of falsely falsifying a faithful candidate, under optional stopping and adaptive budget routing; FDR control of the rejected set; and — for the model one finally accepts — an ε-certification theorem whose fresh-oracle e-process bounds the probability of ever certifying a candidate that disagrees with the specification on at least an ε-fraction of enumerable instances. On real CP-SAT probes no faithful candidate was falsified across every operating level tested, nine of ten injected faults were caught within sub-second budgets, and independently auditing every issued certificate placed all 289 error intervals entirely below ε, with none above and none inconclusive — a cleaner outcome than the $\alpha = 0.05$ per-run budget requires. The broader claim is methodological: solver-in-the-loop verification of generative models is, at bottom, a sequential inference problem — and once it is treated as one, the difference between "we found no bugs" and "we certify at level α" becomes a theorem rather than a hope.

Statements and Declarations

Funding. The authors did not receive support from any organization for the submitted work.

Competing interests. The authors have no competing interests to declare that are relevant to the content of this article.

Data availability. All code, calibration data, experimental results, and cached LLM generations are available in the supplementary material.

Use of large language models. LLMs are the object of study; their use in the experiments (models, prompting, decoding parameters, caching) is documented in Section **??**.

Appendix A Validity proofs for the probe library

Throughout, $\mathcal{X}(I)$ denotes the specification's feasible set on instance I, f_I its objective, and $v^*(I) = \text{opt}_{x\in\mathcal{X}(I)} f_I(x)$. Because tier-B probes solve the *candidate* exactly on both sides, and a faithful candidate satisfies $v_m \equiv v^*$, it suffices to prove each relation for v^*. Three lemmas cover all but one relation.

Lemma 8 (Relaxation) *If $\mathcal{X}(I) \subseteq \mathcal{X}(I')$ and $f_{I'} = f_I$, then $v^*(I') \geq v^*(I)$ for maximization and $v^*(I') \leq v^*(I)$ for minimization.*

Proof The optimum over a superset is no worse: any x optimal for I is feasible for I' with the same objective value. □

Lemma 9 (Objective dominance) *If $\mathcal{X}(I') = \mathcal{X}(I)$ and $f_{I'}(x) \geq f_I(x)$ for all $x \in \mathcal{X}(I)$, then $v^*(I') \geq v^*(I)$, for maximization and minimization alike.*

Proof For maximization, dominate the maximizer of I; for minimization, $\min_x f_{I'}(x) \geq \min_x f_I(x)$ since every point's value rose. □

Lemma 10 (Symmetry) *If π is a bijection of the decision index set and I' is I with all indexed data consistently relabeled by π, then $x \mapsto x \circ \pi^{-1}$ is a bijection $\mathcal{X}(I) \to \mathcal{X}(I')$ preserving objective values; hence $v^*(I') = v^*(I)$.*

Proof Every constraint and objective term of the specification refers to indices only through the relabeled data, so feasibility and value are preserved in both directions. □

Knapsack with conflicts (max).

Capacity increase relaxes the single budget constraint (Lemma 8: ↑). Deleting a conflict pair removes a constraint (Lemma 8: ↑). Item permutation, with the conflict list relabeled by the same permutation, is Lemma 10.

Set cover (min).

Deleting an element removes one covering constraint (Lemma 8: ↓). Raising one set's cost is Lemma 9 on a minimization (↑). Permuting the set collection (costs and coverage lists together) is Lemma 10.

Graph colouring (min).

The specification minimizes the number of colours; the implemented objective, $1 + \max_i c_i$ over proper colourings with colours in $\{0, \ldots, n-1\}$, agrees with it at the optimum because any proper colouring with k distinct colours can be renumbered onto $\{0, \ldots, k-1\}$ without violating properness. Edge deletion removes a disequality constraint (Lemma 8: ↓); edge addition adds one, shrinking the feasible set (Lemma 8 applied with the roles of I, I' exchanged: ↑); vertex relabeling is Lemma 10.

Generalized assignment (min).

Raising one machine's capacity relaxes its load constraint (Lemma 8: ↓). Raising one cost entry is Lemma 9 (↑). Task permutation (sizes and cost rows together) is Lemma 10.

On-time job selection (max).

This is the delicate family. The specification selects $S \subseteq \{1, \ldots, n\}$ maximizing $\sum_{j \in S} w_j$ subject to *schedulability*: some single-machine order completes every job of S by its deadline. The classical characterization states that S is schedulable iff processing S in nondecreasing-deadline (EDD) order is feasible, iff

$$\textstyle\sum_{i \in S:\, d_i \le d_j} p_i \;\le\; d_j \qquad \text{for every } j \in S. \tag{A1}$$

Weight increase is Lemma 9 (↑). *Processing-time decrease*: in any fixed order every completion time weakly decreases, so every schedulable S remains schedulable; the feasible set (in the schedulability sense) grows, and Lemma 8 gives ↑. *Deadline increase* $d_k \mapsto d_k + \delta$: note that the index sets in (A1) change with the deadline order, so monotonicity of the inequality system is not syntactic. Argue at the schedulability level instead: fix a schedule witnessing feasibility of S under the old deadlines; completion times are unchanged and deadlines are pointwise no smaller, so the same schedule witnesses feasibility under the new deadlines. Hence every schedulable set remains schedulable, and Lemma 8 gives ↑; the candidate, if faithful, satisfies (A1) on both sides because (A1) is equivalent to schedulability instance by instance. Job permutation (processing times, deadlines, and weights together) is Lemma 10. □

Appendix B Natural-language specifications and prompts

For reproducibility, the exact specification texts and the prompt template of the end-to-end study (Section ??). Candidates were requested as a single Python function `build_model(params)` returning a CPMpy model with the objective set, under three style hints — *direct* ("Write the model directly.", temperature 0.3), *stepwise* ("First restate the decision variables, constraints and objective in one short paragraph each, then write the code.", 0.7), and *expert* ("You are an expert constraint-programming modeller known for scrupulously faithful formalizations. Write the model.", 0.5).

Template.

```
Translate the following optimization problem into a CPMpy model.
{SPECIFICATION}
{STYLE HINT}
Requirements: provide a single Python function build_model(params) that
returns a cpmpy.Model with the objective set (use model.maximize(...) or
model.minimize(...)); use only the cpmpy library (imported as cp) and the
Python standard library; the function must work for ANY params dict matching
the schema; reply with ONLY a Python code block.
```

Specification texts (abridged schema lines).

Knapsack with conflicts: *"You must select a subset of n items. Item i has weight $w[i]$ and value $v[i]$. The total weight of selected items must not exceed the capacity cap. Some pairs of items conflict: for each pair (i, j) in the list conflicts, items i and j must not both be selected. Maximize the total value of the selected items."* Set cover: *"There are nelem elements, numbered $0..\text{nelem} - 1$, and a collection of sets. Set s covers the elements listed in sets$[s]$ and costs cost$[s]$. Choose a sub-collection of sets so that every element is covered by at least one chosen set. Minimize the total cost of the chosen sets."* Graph colouring: *"Color the n vertices of an undirected graph, using color indices from 0 to $n-1$, so that the two endpoints of every edge receive different colors. Minimize the number of colors used. IMPORTANT convention: the objective value must equal 1 plus the maximum color index used."* Generalized assignment: *"Assign each of T tasks to exactly one of M machines. Task t has size $p[t]$; machine m has capacity cap$[m]$: the total size of tasks assigned to machine m must not exceed cap$[m]$. Assigning task t to machine m costs cost$[t][m]$. Minimize the total assignment cost."* On-time job selection: *"A single machine processes jobs one at a time. Job j has processing time $p[j]$, deadline $d[j]$ and weight $wt[j]$. Select a subset of jobs that* *can ALL be completed by their deadlines when the selected jobs are processed in non-decreasing deadline order (equivalently: a selected set is feasible if and only if, for every selected job j, the sum of processing times of selected jobs i with $d[i] \le d[j]$ is at most $d[j]$). Maximize the total weight of the selected jobs."*

The colouring convention sentence is deliberate: without it, "minimize the number of colors" admits the objective $|\{c_i\}|$ as a faithful reading, and the two readings differ only off-optimum — the template for the planned ambiguous-specification study (Section 10).

Appendix C The mutation operator catalogue

Each operator is a deterministic function of instance parameters, drawn once per candidate and applied identically to every instance the candidate ever sees (a wrong *model*, not a wrong instance). Descriptors are sampled uniformly over the applicable (operator, target) pairs of Table 2; positional indices are drawn once and taken modulo the current instance's dimension, which is what makes them instance-generic.

scalar shift a named scalar bound $b \mapsto \max(1, b + \delta)$, $\delta \in \{\pm1, \pm2, \pm3\}$ (misread constant).
vector entry one positionally selected entry of a named numeric vector shifted by δ as above (misread datum).
vector rotation a named vector rotated by one position — the data-wiring misalignment in which item i receives item $(i+1)$'s datum; the mutation class that only the permutation MR and value-level oracles can see.
row drop one positionally selected structural row (conflict pair, edge, coverage list) deleted (missed constraint).
family half-drop the second half of a structural family deleted (systematically missed constraint class).
matrix entry one positionally selected entry of a cost matrix shifted by δ.
operator flip one constraint family's comparison direction inverted at the builder level ($\le \leftrightarrow \ge$; for colouring, disequality to equality) — the class that breaks monotone MRs and is otherwise invisible to agreement checks when the flipped model is infeasible-prone.

Screening. A drawn mutant is kept only if its exact optimum differs from the specification's brute-forced optimum on at least one of six random micro-instances (up to 40 redraws); this discards semantically equivalent mutants, the standard practice in mutation testing, and guarantees ground-truth labels for every experiment in Section 7.

Appendix D Extended numerical tables

For re-derivation without running code, the full operating-range table behind Figure 5 (100 replications per level, Kelly routing; kill cost is mean measured probe time to reject a mutant):

Table D1 Full α-sweep. These runs exercise the *screening* stage only (no certification phase), which is why detection at $\alpha = 0.05$ here (0.920) sits below the corresponding row of Table 4 (0.958), where fresh-oracle certification probes falsify additional mutants.

α	false rej.	n faithful	detection	n mutants	kill cost
0.01	0.0000	285	0.873	315	21.0 ms
0.02	0.0000	317	0.908	283	21.1 ms
0.05	0.0000	287	0.920	313	14.0 ms
0.10	0.0000	309	0.921	291	14.3 ms
0.20	0.0000	280	0.956	320	13.2 ms

The kill-cost column is Proposition 5 measured: tightening the guarantee from $\alpha = 0.20$ to $\alpha = 0.01$ — a 20× stronger error bound — raises the mean probe cost of a rejection from ∼13 ms to ∼21 ms, a factor of ∼1.6, in line with the predicted $\log(1/0.01)/\log(1/0.20) \approx 2.9$ ratio of log-thresholds attenuated by hard-certificate rejections whose cost is α-independent.

Calibrated betting multipliers implied by Table 3 (equation (1)): tier A alarm ×30.9, pass ×0.535; tier C alarm ×3.00, pass ×0.607; tier B pass ×1, alarm $= \infty$. Rejection threshold at $\alpha = 0.05$: wealth 20, i.e. log-threshold 3.00.

Appendix E Complete generation record

Every LLM generation across both studies (main: Section ??; ambig: Section 7.10), in request order. "free" is the free-tier routed model; "nemotron" is `nvidia/nemotron-3-ultra-550b`; error classes abbreviate the exception type raised at the intake gate (AttrErr: `AttributeError`, ModErr: `ModuleNotFoundError`, TypeErr: `TypeError`, SynErr: `SyntaxError`); the raw responses ship in the supplementary cache.

References

[1] Michailidis, K., Tsouros, D., Guns, T.: Constraint modelling with LLMs using in-context learning. In: 30th International Conference on Principles and Practice of Constraint Programming (CP 2024). LIPIcs, vol. 307 (2024). https://doi.org/10.4230/LIPIcs.CP.2024.20

[2] Shi, W., Liu, M., Zhang, W., Shi, L., Jia, F., Ma, F., Zhang, J.: ConstraintLLM: A neuro-symbolic framework for industrial-level constraint programming. In: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP) (2025). https://arxiv.org/abs/2510.05774

[3] Szeider, S.: CP-Agent: Agentic constraint programming (2025). https://arxiv.org/abs/2508.07468

[4] Ramdas, A., Grünwald, P., Vovk, V., Shafer, G.: Game-theoretic statistics and

Table E2 All 48 generations.

Study	Family	Model	Style	Intake	Outcome / error
main	knapsack_conflicts	free	direct	fail	`AttrErr`
main	knapsack_conflicts	free	stepwise	ok	faithful
main	knapsack_conflicts	free	expert	ok	faithful
main	knapsack_conflicts	nemotron	direct	ok	faithful
main	knapsack_conflicts	nemotron	stepwise	ok	faithful
main	knapsack_conflicts	nemotron	expert	ok	faithful
main	set_cover	free	direct	fail	`AttrErr`
main	set_cover	free	stepwise	ok	faithful
main	set_cover	free	expert	fail	`AttrErr`
main	set_cover	nemotron	direct	ok	faithful
main	set_cover	nemotron	stepwise	ok	faithful
main	set_cover	nemotron	expert	ok	faithful
main	coloring	free	direct	ok	faithful
main	coloring	free	stepwise	ok	faithful
main	coloring	free	expert	fail	`TypeErr`
main	coloring	nemotron	direct	ok	faithful
main	coloring	nemotron	stepwise	ok	faithful
main	coloring	nemotron	expert	ok	faithful
main	gen_assignment	free	direct	fail	`AttrErr`
main	gen_assignment	free	stepwise	fail	`AttrErr`
main	gen_assignment	free	expert	fail	`AttrErr`
main	gen_assignment	nemotron	direct	ok	faithful
main	gen_assignment	nemotron	stepwise	ok	faithful
main	gen_assignment	nemotron	expert	ok	faithful
main	ontime_selection	free	direct	fail	`ModErr`
main	ontime_selection	free	stepwise	fail	`AttrErr`
main	ontime_selection	free	expert	ok	faithful
main	ontime_selection	nemotron	direct	ok	faithful
main	ontime_selection	nemotron	stepwise	ok	faithful
main	ontime_selection	nemotron	expert	fail	`AttrErr`
ambig	coloring	free	direct	fail	`SynErr`
ambig	coloring	free	stepwise	fail	`ModErr`
ambig	coloring	free	expert	fail	`AttrErr`
ambig	coloring	nemotron	direct	ok	canonical
ambig	coloring	nemotron	stepwise	fail	`AttrErr`
ambig	coloring	nemotron	expert	ok	canonical
ambig	knapsack_conflicts	free	direct	fail	no code block
ambig	knapsack_conflicts	free	stepwise	fail	`AttrErr`
ambig	knapsack_conflicts	free	expert	fail	no code block
ambig	knapsack_conflicts	nemotron	direct	ok	canonical
ambig	knapsack_conflicts	nemotron	stepwise	ok	canonical
ambig	knapsack_conflicts	nemotron	expert	ok	canonical
ambig	ontime_selection	free	direct	ok	canonical
ambig	ontime_selection	free	stepwise	ok	canonical
ambig	ontime_selection	free	expert	fail	`AttrErr`
ambig	ontime_selection	nemotron	direct	ok	canonical
ambig	ontime_selection	nemotron	stepwise	ok	canonical
ambig	ontime_selection	nemotron	expert	ok	canonical

safe anytime-valid inference. Statistical Science **38**(4), 576–601 (2023) https://doi.org/10.1214/23-STS894

[5] Shafer, G.: Testing by betting: A strategy for statistical and scientific communication. Journal of the Royal Statistical Society: Series A **184**(2), 407–431 (2021) https://doi.org/10.1111/rssa.12647

[6] Grünwald, P., Heide, R., Koolen, W.M.: Safe testing. Journal of the Royal

Statistical Society Series B: Statistical Methodology **86**(5), 1091–1128 (2024) https://doi.org/10.1093/jrsssb/qkae011

[7] Ville, J.: Étude Critique de la Notion de Collectif. Gauthier-Villars, Paris (1939)

[8] Kelly, J.L.: A new interpretation of information rate. Bell System Technical Journal **35**(4), 917–926 (1956) https://doi.org/10.1002/j.1538-7305.1956.tb03809.x

[9] Wang, R., Ramdas, A.: False discovery rate control with e-values. Journal of the Royal Statistical Society Series B: Statistical Methodology **84**(3), 822–852 (2022) https://doi.org/10.1111/rssb.12489

[10] Guns, T.: Increasing modeling language convenience with a universal n-dimensional array: CPpy as python-embedded example. In: Proceedings of the 18th Workshop on Constraint Modelling and Reformulation (ModRef) (2019)

[11] Perron, L., Furnon, V.: OR-Tools CP-SAT solver. Google, https://developers.google.com/optimization (2024)

[12] Michailidis, K., Tsouros, D., Guns, T.: DCP-Bench-Open: Evaluating LLMs for constraint modelling of discrete combinatorial problems. CP-Bench version accepted at ECAI 2025 (2025). https://arxiv.org/abs/2506.06052

[13] Szeider, S.: MCP-Solver: Integrating language models with constraint programming systems (2025). https://arxiv.org/abs/2501.00539

[14] Chen, T.Y., Kuo, F.-C., Liu, H., Poon, P.-L., Towey, D., Tse, T.H., Zhou, Z.Q.: Metamorphic testing: A review of challenges and opportunities. ACM Computing Surveys **51**(1), 4–1427 (2018) https://doi.org/10.1145/3143561

[15] DeMillo, R.A., Lipton, R.J., Sayward, F.G.: Hints on test data selection: Help for the practicing programmer. Computer **11**(4), 34–41 (1978) https://doi.org/10.1109/C-M.1978.218136

[16] Jia, Y., Harman, M.: An analysis and survey of the development of mutation testing. IEEE Transactions on Software Engineering **37**(5), 649–678 (2011) https://doi.org/10.1109/TSE.2010.62

[17] Li, H., Hai, M.: Falsification-based verification of LLM-generated optimization models: Sound test batteries and their detection limits (2026). https://arxiv.org/abs/2607.16646

[18] Zadorojniy, A., Wasserkrug, S., Farchi, E.: An agent-based framework for the automatic validation of mathematical optimization models (2025). https://arxiv.org/abs/2511.16383

[19] Vovk, V., Wang, R.: E-values: Calibration, combination and applications. The Annals of Statistics **49**(3) (2021) https://doi.org/10.1214/20-AOS2020

[20] Waudby-Smith, I., Ramdas, A.: Estimating means of bounded random variables by betting. Journal of the Royal Statistical Society Series B: Statistical Methodology **86**(1), 1–27 (2024) https://doi.org/10.1093/jrsssb/qkad009

[21] Clopper, C.J., Pearson, E.S.: The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika **26**(4), 404–413 (1934) https://doi.org/10.1093/biomet/26.4.404

[22] Freund, Y., Schapire, R.E.: A decision-theoretic generalization of on-line learning and an application to boosting. Journal of Computer and System Sciences **55**(1), 119–139 (1997) https://doi.org/10.1006/jcss.1997.1504

[23] Cesa-Bianchi, N., Lugosi, G.: Prediction, Learning, and Games. Cambridge University Press, Cambridge (2006). https://doi.org/10.1017/CBO9780511546921